

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech.

Branch: Electronics Engineering/Electronics and Communication Engineering

/Electronics and Tele-communication Engineering

Semester :IV

Subject Code & Name: (BTBS404) Probability Theory and Random Processes

Max Marks: 60

Date:23/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following. 12

- | | | |
|---|-----|---|
| A) State and prove the Law of Total Probability. | CO3 | 6 |
| B) State and prove the Bayes' Theorem. | CO3 | 6 |
| C) A person has undertaken a mining job. The probabilities of completion of the job on time with and without rain are 0.42 and 0.90 respectively. If the probability that it will rain is 0.45, then determine the probability that the mining job will be completed on time. | CO3 | 6 |

Q.2 Solve Any Two of the following. 12

- | | | |
|--|-----|---|
| A) If the random variable X takes the values 1, 2, 3 and 4 such that $2P(X = 1) = 3P(X = 2) = P(X = 3) = 5P(X = 4)$, find the probability distribution and cumulative distribution function of X. | CO1 | 6 |
| B) Define CDF and state its properties. | CO1 | 6 |
| C) If the density function of a continuous random variable X is given by | CO3 | 6 |
- $$f_X(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ 3a - ax, & 2 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$
- i. Find the value of a
 - ii. Find the CDF of X

Q. 3 Solve Any Two of the following. 12

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|---|-----|---|
| A) Define covariance matrix and state its properties. | CO4 | 6 |
|---|-----|---|

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|---|-----|---|
| B) Explain Normal Distribution with mathematical expressions. | CO3 | 6 |
| C) A die is thrown 6 times. If 'getting an odd number' is a success, what is the probability of | CO3 | 6 |
| (i) 5 successes | | |
| (ii) at least 5 success | | |
| (iii) at most 5 successes | | |

Q.4 Solve Any Two of the following. 12

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|--|-----|---|
| A) Write a short note on Law of Large Numbers. | CO3 | 6 |
| B) State the Central limit theorem and its significance. | CO2 | 6 |
| C) If 3% of electronic units manufactured by a company are defective. Find the probability that in a sample of 200 units, less than 2 bulbs are defective. | CO1 | 6 |

Q. 5 Solve Any Two of the following. 12

- | | | |
|--|-----|---|
| A) Define Power Spectral Density and state its properties. | CO2 | 6 |
| B) Write a short note on Ergodicity. | CO2 | 6 |
| C) Given that the autocorrelation function for a stationary ergodic process with no periodic components is | CO2 | 6 |

$$R_{xx}(\tau) = 25 + \frac{4}{1 + 6\tau^2}$$

Find the mean value and variance of the process $\{X(t)\}$.

***** End *****